

ST512-SP26-Homework-5

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Problem 1

a

Source	df	SS	MS	F-value	p-value
Factor A	4	375	93.75	27.79	3.66e-14
Factor B	2	462	231.0	68.49	0
AxB Interaction	8	38	4.75	1.40	0.210
Error	75	253	3.373	–	–
Total	89	1128	–	–	–

b

Based on the ANOVA table, the interaction factor has a p-value of 0.210. Using a 0.05 significance level, we fail to reject the null hypothesis. Therefore, there is NOT a significant interaction between factor A and factor B.

c

Based on the ANOVA table, for Factor A the p-value was calculated to be 3.66e-14. Using a significance value of 0.05, we reject the null hypothesis. Therefore, there is a significant difference in mean response across the levels of factor A.

d

Based on the ANOVA table, for Factor B the p-value was calculated to be 0. Using a significance value of 0.05, we reject the null hypothesis. Therefore, there is a significant difference in mean response across the levels of factor B.

Problem 2

a

a=4, b=3, n=3, N=36, SSE=720

df (error)= 24, MSE=720/24=30.0

No interaction → each factor A mean averages over b*n=9

Tukey MSD:

$$q_{\alpha,4,24} * \sqrt{MSE/n_A} = 3.90 * \sqrt{30/9} = 7.12$$

Bonferroni MSD:

$$t_{\frac{\alpha}{2C},24} * \sqrt{2MSE/n_A} = 2.875 * \sqrt{60/9} = 7.42$$

b

Interaction → compare Factor A within one level of Factor B, n=3

Tukey MSD:

$$q_{\alpha,4,24} * \sqrt{MSE/n_A} = 3.90 * \sqrt{30/3} = 12.33$$

Bonferroni MSD:

$$t_{\frac{\alpha}{2C},24} * \sqrt{2MSE/n_A} = 2.875 * \sqrt{60/3} = 12.86$$

Problem 3

a

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

i=1,2,3 (variety); j=1,2,3(temp); k=1,...,8

Y_{ijk} = total dry weight of plant material

μ = overall mean

α_i = effect of variety i, $\sum \alpha_i = 0$

β_j = effect temperature j, $\sum \beta_j = 0$

$(\alpha\beta)_{ij}$ = interaction effect subject to $\sum (\alpha\beta)_{ij} = 0$

ϵ_{ijk} = random error $\sim N(0, \sigma^2)$

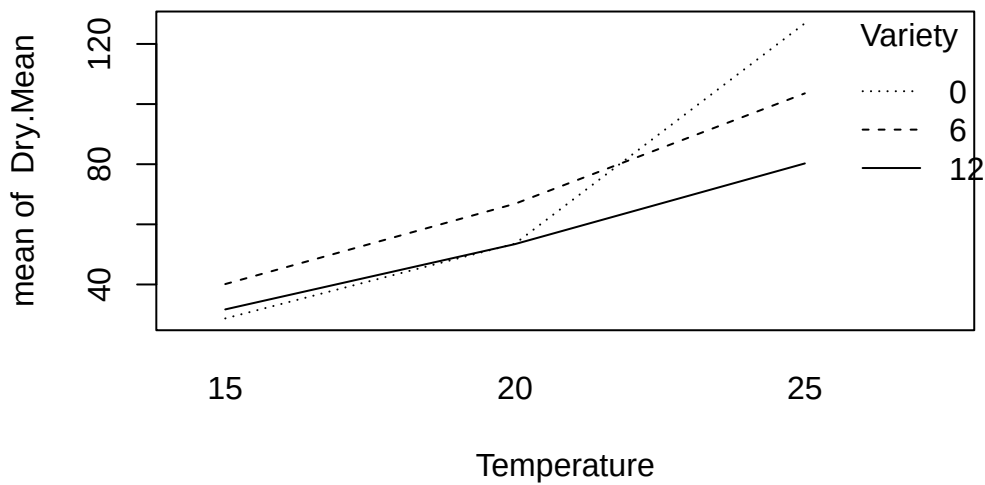
assumptions: independence, normality, equal variance

b

Source	df	SS	MS	F-value	p-value
model	8	71692.11	8961.51	189.01	<0.0001
Variety	2	3325.78	1662.89	35.07	<0.0001
Temperature	2	61549.77	30774.86	649.08	<0.0001
Var x Temp	4	6817.6	1704.4	35.95	<0.0001
Error	63	2987.02	47.413	—	—
Total	79	146372.28	—	—	—

c

[1] "Variety" "Temperature" "Dry.Mean"



All three varieties increase with temperature, but at different rates. The lines cross, indicating a significant interaction. Variety 0 is lowest at 15°C but highest at 25°C.

d

$$H_o : all(\alpha\beta)_{ij} = 0$$

$$H_A : atleastone(\alpha\beta)_{ij} \neq 0$$

$$\text{F-value: } \frac{MS_{AB}}{MSE} = 1704.4/47.413 = 35.95$$

$$1-\text{pf}(35.94, 4, 63) = \text{p value} = 1.33\text{e-}15$$

There is a significant interaction between Variety and Temperature.

R code

problem 1

```
1-pf(27.79,4,75)
```

```
[1] 3.663736e-14
```

```
1-pf(68.49, 2,75)
```

```
[1] 0
```

```
1-pf(1.40,8,75)
```

```
[1] 0.2105261
```

problem 2

```
qtukey(.95,4,24)
```

```
[1] 3.901262
```

```
qt(0.0041667,24)
```

```
[1] -2.875091
```

problem 3

```
1-pf(35.95,4,63)
```

```
[1] 1.332268e-15
```

```
plant_data<-read.csv("plant_data.csv",header=TRUE)  
attach(plant_data)
```

The following objects are masked `_by_` .GlobalEnv:

Temperature, Variety

The following objects are masked from `plant_data` (`pos = 3`):

Dry.Mean, Temperature, Variety

```
names(plant_data)
```

```
[1] "Variety"      "Temperature" "Dry.Mean"
```

```
Variety<-factor(Variety)  
Temperature<-factor(Temperature)  
interaction.plot(Temperature, Variety, Dry.Mean)
```

